

Integrating Differentiable Ray Tracing and Neural Network for Advanced DifferentiableRF Modeling

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Abstract

This research proposal explores differentiable Radio Frequency (RF) modeling, an emerging approach for optimizing wireless communication systems and environmental simulations. The key idea is to leverage automatic differentiation to propagate gradients from RF signal data back to scene parameters, enabling gradient-based optimization algorithms. In this proposal, we propose a hybrid approach that combines both differentiable simulators and neural networks, aiming to leverage the strengths of each. By analyzing the challenges in combining these methods, we identify several promising directions for exploration. Through this work, we aim to advance the state of differentiable RF modeling, contributing to more effective and accurate tools for future wireless communication networks.

1 Introduction

Differentiable Radio Frequency (RF) Modeling has recently attracted significant interest in wireless communication, sensing, and simulation. [9][5][6] It enables gradient-based optimization via automatic differentiation of environmental parameters, such as material properties, providing an indispensable framework for solving inverse electromagnetic(EM) rendering problems. This differentiable paradigm unlocks novel approaches for wireless network performance optimization, establishing itself as a cornerstone technology for future wireless networks.

Conventional RF modeling formulates wireless propagation as parametric functions $f(\theta)$ where θ encompasses environmental geometry, material properties and transceiver or receiver configurations to multi-path channel information [18]. The differentiable paradigm uniquely leverages automatic differentiation to compute gradients of channel information with respect to the scene parameters. By implementing automatic differentiation through frameworks like PyTorch, these techniques compute gradients via back-propagation without relying on error-prone finite-difference approximations.

As illustrated in figure 1, there are two main paradigms in Differentiable RF Modeling:

Differentiable Simulator: This method includes various techniques, with ray tracing being a commonly used component to simulate the propagation of electromagnetic waves. Based on the similarity between light and electromagnetic wave propagation, ray tracing calculates different kinds of interactions of electromagnetic waves with objects in the environment, including reflection, refraction, and scattering[?]. By incorporating automatic differentiation into the ray tracing process, it allows for the computation of gradients with respect to simulation parameters.

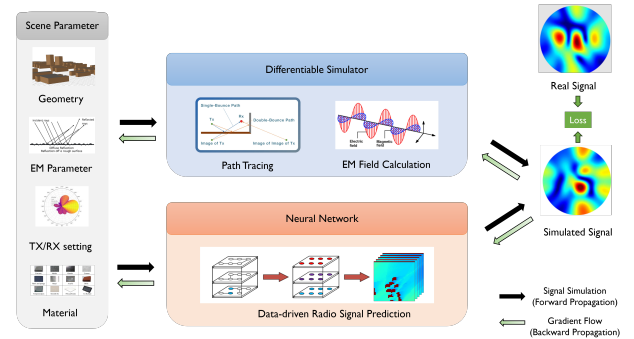


Figure 1: The framework for Differentiable Radio Propagation Modeling, where the differentiable simulator or neural network maps environment parameters to corresponding signal outcomes during forward propagation, and computes the derivatives of the signal with respect to scene parameters during backward propagation.

Neural Network: This method leverages neural networks to model radio propagation, treating it as a function that can be learned through data-driven approaches. Utilizing architectures such as Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), these models are trained on diverse datasets representing various scene parameters. This allows the networks to capture complex interactions between signals and the environment. [13] What's more, the differentiability of these models enables gradient computation and optimization, which is achieved through the backpropagation process.

Each paradigm offers distinct advantages. Differentiable simulators, particularly those using ray tracing, provide high accuracy and clear interpretability due to their strong physical foundation. This makes it easier for researchers to analyze experimental phenomena and results, offering a robust framework for understanding the underlying physics of radio propagation. Neural network-based methods, on the other hand, demonstrate superior performance in fast inference and strong fitting capabilities for real-world signals. Their ability to quickly process data makes them suitable for real-time applications, enable them to accurately model and predict complex signal behaviors.

Despite the potential demonstrated by both methods, each approach faces its limitations. For differentiable simulators, the accuracy of simulation results heavily depends on the precision of input scene parameters. This is often impractical in many scenarios, as it requires detailed EM measurements for every object in the

environment. Although gradient-based calibration can partially address this issue, these methods often exhibit significant simulation-to-reality gaps due to simplified models that cannot fully capture the complexity of real-world objects, such as irregular shapes and heterogeneous materials.

Neural networks, on the other hand, inevitably encounter issues related to dataset scarcity and generalization. Due to the absence of comprehensive multi-scenario datasets containing diverse frequency configurations, environmental condition, the neural networks often fit the electromagnetic characteristics of specific environments on small datasets, but struggle to generalize across different frequencies, environments, and transmitter-receiver positions. Furthermore, the lack of datasets critically impedes the development of foundation models capable of learning generalized electromagnetic scattering patterns essential for robust real-world deployment.

2 Motivation

The motivation behind this research is to combine differentiable simulators with neural networks to design improved differentiable RF modeling. This approach is to some extent similar to Physics-informed AI, it leverages the strengths of both methodologies: differentiable simulators provide a physically grounded framework, while neural networks are dedicated to learn from data. Additionally, due to the chain rule in backpropagation, the integration of the differentiable simulators with neural networks retains the differentiability property, allowing gradients to be computed efficiently throughout the whole model. This hybrid approach offers several key advantages:

Data Efficiency: Acquiring high-quality labeled datasets remains a major bottleneck in RF modeling, where data collection is costly and time-consuming. Incorporating physical knowledge as strong priors within the learning framework substantially alleviates this requirement by constraining the solution space to physically plausible behaviors. This integration enables the model to achieve reliable performance even with limited or incomplete datasets, thereby increasing its applicability to real-world scenarios where extensive annotations are impractical or unavailable.

Model Interpretability: Deep neural networks are often criticized for their lack of transparency, making it difficult to interpret the rationale behind their predictions. Embedding physical laws within the model architecture enforces learning of well-defined features grounded in real-world phenomena, thus improving interpretability. Moreover, by focusing on physically meaningful representations, the model avoids fitting overly complex or abstract features, which facilitates faster and more stable convergence during training. This leads to improved training efficiency and robustness, critical factors in practical RF modeling applications where both interpretability and reliable performance are essential.

Physics-Guided Fusion: Traditional neural network approaches often face significant challenges when integrating heterogeneous data sources such as point clouds or images. These methods typically require complex preprocessing steps to make disparate data

compatible. In contrast, differentiable simulators leverage fundamental physical principles, specifically electromagnetic wave propagation laws, to provide a unified and physics-informed framework for data fusion. This enables natural integration of multiple data modalities while preserving physical consistency. Consequently, the resulting models exhibit enhanced robustness and superior generalization capabilities in complex and previously unseen environments, where data distributions may vary substantially from training conditions.

3 Related Work and Challenges

There are a lot of work has been conducted on neural rendering in the area of computer graphics, including well-known methods such as NeRF[14] and 3DGS[12]. In recent years, researchers have begun to draw inspiration from these approaches and apply them to RF modeling and one of the most significant works is NERF2[19]. NERF2 introduces a novel Neural Radio-Frequency Radiance Field, it conceptualizes a continuous, volumetric scene function to interpret RF signal propagation. Specifically, this method defines the environment as multiple voxels, where the attenuation and radiance of each voxel influence the corresponding signal's amplitude and phase. These parameters are encoded using a Multi-Layer Perceptron (MLP) and are further refined through turbo-learning, which combines synthetic and actual datasets for enhanced training.

As NERF2 serves as an excellent example of "Neural Rendering in RF Modeling", but there are still many areas for further improvement. For example, in NERF2, every pixel in the space requires training, but the air-like regions, which do not significantly contribute to attenuation, could be excluded from consideration. Similar adjustments would significantly reduce the data volume requirements and training time.

Several other works have also addressed this issue. Sionna RT performs standard ray tracing, including reflection and refraction, and uses an MLP to store environmental information about buildings, calibrating ray tracing accordingly.[9] In [3], the research utilizes Signed Distance Fields (SDF) for RF ray tracing, ensuring differentiability, and uses an MLP to learn distance and attenuation. RFCanvas [4] leverages existing visual information to help construct Tensorial Fields and learn attenuation and radiance. WRF-GS[17] represents wireless channels using 3D Gaussian Splatting, taking advantage of the differentiability of Gaussian Splatting for learning.

Environmental Representation: In computer graphics, various data representations have been explored for modeling environments, such as meshes, point clouds, images, and implicit.[16] Each representation comes with its own set of advantages and limitations: for instance, mesh-based models are computationally efficient but are typically limited in capturing fine details of surface roughness or complex geometries; volumetric rendering, while capable of representing more intricate material properties, demands significantly more computational resources. When transitioning these representations to RF modeling, the challenge lies in selecting or developing environment representations that can effectively capture the complex electromagnetic phenomena, such as reflection, refraction, and diffraction[1], while maintaining computational efficiency. The need to balance physical fidelity with computational

load becomes particularly pronounced in large-scale or dynamic environments.

Architecture Design: A key challenge in RF modeling is designing architectures that accurately simulate RF signal propagation in diverse environments. The integration of physical domain knowledge with data-driven methods introduces additional complexity, requiring a clear distinction between components governed by physical laws, such as path loss, and those learned from data, like diffraction and small-scale fading. The model's design becomes particularly challenging as it must seamlessly combine these two elements—ensuring physical components maintain simulation accuracy while allowing data-driven components to generalize across varied environments. Furthermore, the scarcity of high-quality labeled data exacerbates this challenge. To mitigate this, techniques such as data augmentation may be necessary to enrich the training dataset and reduce dependence on labeled data.

Computational Efficiency and Acceleration: This part focuses more on the practical challenges encountered in detailed implementation. Differentiable simulations (e.g., ray tracing) face high computational complexity in large-scale environments, which intensifies significantly when integrating physics-based models with data-driven methods like neural networks. Simulators must simultaneously compute precise physical processes (e.g., electromagnetic wave propagation) and handle the computational overhead of data-driven components. Balancing physical accuracy with data-driven learning requires careful optimization of computational resources and model architecture, while large-scale data and cross-domain computations further amplify the burden. Efficiently unifying these physics-based and data-driven approaches is therefore critical for achieving real-world feasibility without sacrificing accuracy.

4 Methodology

In this section, we propose three distinct approaches to address the challenges of accurate and efficient RF signal propagation modeling in complex environments. Each approach combines neural networks with differentiable simulations, as illustrated in Figure 2.

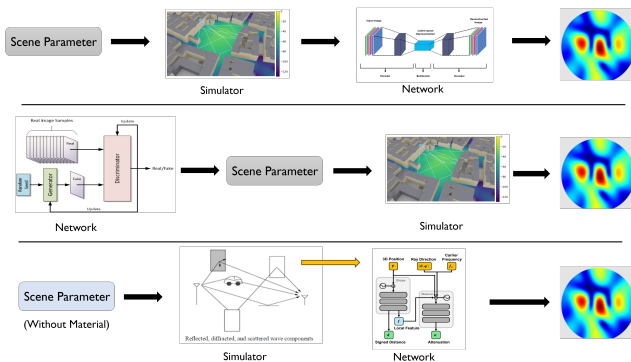


Figure 2: Illustration of three approaches for integrating neural networks with differentiable simulators to improve RF modeling.

The first approach focuses on using a neural network to reduce the discrepancy between simulated and real-world RF propagation results. Initially, traditional simulation models, such as ray tracing, are used to predict electromagnetic wave behavior in a given environment. However, these simulations may fail to capture all the nuances of real-world environments. To improve the simulation's accuracy, a neural network is trained to minimize the difference between the simulated predictions and actual measurements. The network learns the residual errors and provides corrections to the simulated outputs, thus improving their fidelity without requiring additional, computationally expensive simulations. In this approach, Autoencoders (AE)[2] are employed for feature extraction and dimensionality reduction. AE networks compress the input data into a lower-dimensional latent space, which captures key features of the RF propagation data. This dimensionality reduction helps the network focus on the most relevant features, enhancing the correction process and improving generalization to unseen environments.

The second approach utilizes generative models to synthesize scene parameters, which are then used in ray tracing simulations to predict RF propagation results. Generative models, such as Generative Adversarial Networks (GANs)[7] and Variational Autoencoders (VAEs)[15], are trained to generate realistic scene parameters that describe the environment's geometry, materials, and other relevant features. These generated parameters are then input into a ray tracing engine to simulate the propagation of electromagnetic waves. Generative models like GANs and VAEs offer a powerful way to generate diverse and realistic environments without the need for large labeled datasets or manual data collection. This enables the model to explore a broader range of RF scenarios and improve generalization to new, unseen environments.

The third approach integrates neural networks and ray tracing simulations more deeply. In this approach, the neural network is used to model environmental properties such as material characteristics directly within the simulation process. For example, the network can learn to encode features like surface roughness, reflectivity, and attenuation, which are crucial for accurate wave propagation modeling. During ray tracing, the network queries these learned representations of material properties, enabling dynamic adjustments to the simulation in real time[4]. By embedding the neural network within the ray tracing framework, this hybrid approach enhances the accuracy of electromagnetic wave propagation simulations, particularly in complex environments with diverse materials and geometries. This deeper integration allows the model to combine the strengths of both physical simulations and data-driven learning, improving the overall efficiency and physical consistency of the simulation process.

Each of these methods addresses key challenges in RF signal propagation modeling by combining physical simulations with data-driven approaches, enhancing accuracy, efficiency, and generalization in complex, real-world environments.

5 Progress Overview and Future Work

I have largely completed the implementation of Taichi-RT, a parallelized differentiable ray tracing simulation using the Taichi programming language [11]. Taichi is a high-performance, data-oriented

language designed to streamline high-performance computing, particularly for physics simulations. One of its key advantages is the ability to achieve efficient parallel computing on GPUs with Python-like syntax, eliminating the need for manually written CUDA code. Compared to other frameworks such as TensorFlow and PyTorch, Taichi offers more fine-grained control over computation, making it particularly suited for complex simulation tasks.

Building on this foundation, Taichi-RT can simulate different electromagnetic mechanisms in different scenes efficiently, while leveraging automatic differentiation provided by Taichi[10] to compute the corresponding gradients. Moving forward, I aim to enhance the simulation's capabilities, with a focus on improving both computational accuracy and speed.

Furthermore, we plan to explore the integration of Taichi-RT with neural networks as discussed in the methodology, creating a more unified framework for RF modeling. We will experiment with different types of environmental representations, including point clouds, SDFs, and implicit representations, and explore various neural network architectures to see which one integrates best. Additionally, we will investigate how to better combine the two frameworks based on the knowledge in both wireless communication and neural network. Finally, if conditions permit, I would like to explore the possibility of deploying the above models to actual communication devices, potentially providing a new paradigm for future communication systems.

Furthermore, I will investigate how this unified framework can be extended to Integrated Sensing and Communication (ISAC). The concept of inverse rendering shares many similarities with ISAC[8], especially in the goal of inferring environmental information from observed signals. Just as inverse rendering reconstructs physical properties of a scene from images, ISAC uses RF signals to jointly sense and communicate, extracting environmental data. By merging differentiable ray tracing principles with ISAC, we could develop a comprehensive system capable of both sensing and communication, broadening the applicability of this method in real-world RF scenarios.

6 Conclusion

This research proposal addresses the critical challenge of advancing differentiable RF modeling by proposing a hybrid framework that integrates differentiable simulators with neural networks. We have identified the limitations of existing approaches, the simulation-to-reality gap inherent in differentiable simulators due to imprecise environmental parameters and the generalization challenges faced by neural networks stemming from data scarcity—and propose to overcome these through three synergistic methodologies: residual correction to enhance simulation accuracy, generative modeling to expand data diversity and coverage, and embedded neural representations to achieve deeper physical integration and computational efficiency.

Building on our implemented differentiable ray tracing framework (Taichi-RT), we will refine this foundation through various environmental representations and optimized neural integrations.

By bridging physical simulation with data-driven learning, this research establishes a new paradigm for accurate, efficient, and generalizable RF modeling to advance next-generation wireless systems.

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